

Orientation and exchange statistics in the Golden six-mode conference interferometer

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Abstract

Which features of a one-particle transfer remain observable when its input and output port frames are unframed, oriented, or fully phase-calibrated? We prove that singular values classify an unframed real transfer, that calibrated permanent amplitudes depend on the chosen frames, and that the determinant is the unique minimal-degree polynomial retaining their relative orientation. We realize this hierarchy exactly in a six-mode conference interferometer. For a diagonal path control D_x , its postselected block

$$K_T(x) = Q_{T,-}^T D_x Q_{T,+}$$

has top-exterior amplitude $\det K_T(x)$; its filled-fermion probability is sign-blind. Six modes are the unique nontrivial realized symmetric conference order whose balanced controls all have one exchange spectrum, namely $\{1/5, 4/5, 4/5\}$; the other 44 Boolean controls have zero fermionic amplitude. At every realized higher symmetric-conference order the spectrum varies, but a classical aligned 3-design fixes the exact mean and variance of its purity. The six-mode symmetric- and exterior-cube values are $313/125$ and $16/125$. In the Clebsch realization the six determinants are signed Joubert coordinates on the Segre cubic. Their calibrated signs form a distance-six simplex code. A six-mode compilation and quantitative gates separate three experiments: coherent tomography can infer determinant signs, ordinary photons probe bosonic controls, and direct three-fermion emulation additionally requires a totally antisymmetric three-photon qutrit resource. This is a theory and design-limit analysis, not a report of a built device.

1 Introduction

Let K be a real input–output block of a linear one-particle transfer. Without calibrated port frames, its singular values are intrinsic. After calibration, bosonic collision-free amplitudes are permanents of the framed matrix. After orienting the two port spaces, the top exterior map becomes the signed determinant. These are different levels of information, not three notations for the same observable.

We consider a six-mode transfer obtained from a real symmetric conference matrix C of order six, so that

$$C^T = C, \quad C_{ii} = 0, \quad C_{ij} \in \{\pm 1\}, \quad C^2 = 5I.$$

The two eigenspaces of C , with eigenvalues $\pm\sqrt{5}$, both have dimension three. A diagonal path control couples them through a three-by-three block K . Six coherently marked conference matrices produce six such blocks. The Clebsch source papers construct this Golden operator family from the regular six-point two-graph and prove that its determinants are the six classical Joubert coordinates on the Segre cubic [1, 2]. The present paper builds

on that theorem: it asks what the transfer retains under symmetric and antisymmetric exchange statistics, how its marked determinant signs can be read in a calibrated apparatus, and where a photonic realization stops. The name *Golden* is retained from that series and refers to the paired $\pm\sqrt{5}$ conference eigenspaces; *conference interferometer* names the operational realization studied here.

The paper has two principal results. First, the left–right orbit theorem classifies the information retained by unframed and oriented port spaces and proves that the determinant is the unique minimal-degree polynomial carrying their relative orientation. Second, every balanced Boolean control in the six-mode conference family has squared singular spectrum $\{1/5, 4/5, 4/5\}$. Paper III proves that no higher symmetric conference order has a cut-independent balanced spectrum. At higher orders a credited aligned 3-design instead fixes the switching-class-independent mean and variance of the two-copy purity. The order-six statement is a structural consequence of the signed triangle identity and the $3+3$ conference block equations, not a finite census. It gives exact symmetric, mixed, and exterior exchange-sector benchmarks and the sharp split between twenty invertible and 44 singular Boolean controls. The elementary singular-value classification supplies the invariant language; the substantive combination is its minimal orientation carrier, the uniform conference spectrum, and the calibrated readout boundary.

The orientation mechanism is as follows. The Golden data determine the two three-dimensional port spaces but not ordered orthonormal frames inside them. Changing those frames acts on K from the left and right by orthogonal matrices. The top exterior power is intrinsically a map between the two determinant lines; after both lines are oriented, it becomes the signed scalar $\det K$, while the ordinary filled-fermion probability $|\det K|^2$ is sign-blind. The permanent does not descend through the same port-frame action. Consequently the canonical intrinsic bosonic aggregate used here is not a second signed Segre coordinate: it is a spectral quantity on the symmetric cube.

Theorems 3.1, 3.5, and 3.7 separate the general orientation boundary, the conference classification, and the exact Golden benchmark. Their experimental consequence has three levels. Phase-referenced one-particle tomography infers calibrated determinant signs; ordinary photons measure bosonic controls; direct antisymmetric emulation needs an additional three-qutrit resource. The permanent description of multiphoton scattering is standard [3]; immanants organize the intermediate symmetry sectors [4], and general partial-distinguishability formalisms precede the complementarity identities used here [5]. General boson–fermion complementarity in linear interferometers is due to Jabbour and Cerf [6]; its extension to partially distinguishable particles is due to Robbio, Jabbour, and Cerf [7]. The Clebsch source paper already records the determinant-line interpretation and the failure of the permanent to descend through port-frame changes [2, Section 5]. The present contribution is the general orbit classification and minimal-degree theorem, their exact Golden exchange-sector specialization, and the associated measurement and resource package.

2 The Golden transfer

Here is a reproducible local model of the source object. On the ordered path set $X = \{0, \dots, 5\}$, take

$$C_0 = \begin{pmatrix} 0 & 1 & 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 & -1 & -1 \\ 1 & -1 & 0 & 1 & 1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 1 & -1 & 0 \end{pmatrix}, \quad C_0^2 = 5I.$$

Let $\mathcal{T}$ be the six synthemetic totals, that is, the six labelled one-factorizations of K_X ; below ij denotes the edge $\{i, j\}$. One total is

$$T_0 = \{\{01, 23, 45\}, \{02, 14, 35\}, \{03, 15, 24\}, \\ \{04, 13, 25\}, \{05, 12, 34\}\}.$$

The S_6 -orbit of T_0 has six elements. For each $T \in \mathcal{T}$, choose the lexicographically first permutation π_T carrying T_0 to T , let P_T be its permutation matrix, and put $C_T = P_T C_0 P_T^T$. This is the frozen convention used by the artifact. Transporting the ordered path basis fixes the Hodge and determinant-line orientations, including the signs ε_T below. The Clebsch source theorem proves that this marked family is the conference-operator realization of the six Joubert coordinates [2, Theorem 5.1]; Paper I supplies its upstream two-graph provenance [1].

For one member C_T of this marked conference family, let $Q_{T,+}$ and $Q_{T,-}$ be matrices whose columns are ordered orthonormal frames for the $\pm\sqrt{5}$ -eigenspaces. In the base gauge these frames are obtained by Gram-Schmidt from $P_{\pm}e_0, P_{\pm}e_1, P_{\pm}e_2$, where $P_{\pm} = (I \pm C_0/\sqrt{5})/2$, and are transported by P_T . For $x = (x_0, \dots, x_5)$, put

$$D_x = \text{diag}(x_0, \dots, x_5), \quad K_T(x) = Q_{T,-}^T D_x Q_{T,+}.$$

When $|x_i| = 1$, the six-mode transfer

$$U_T(x) = [Q_{T,+} \ Q_{T,-}]^T D_x [Q_{T,+} \ Q_{T,-}]$$

is orthogonal. Values $|x_i| < 1$ describe a diagonal filter followed by postselection.

The six determinant amplitudes satisfy

$$Z_T(x) = \varepsilon_T 10\sqrt{5} \det K_T(x), \quad \varepsilon_T \in \{\pm 1\}, \quad (1)$$

with signs fixed by the marked orientations. They obey

$$\sum_T Z_T(x) = 0, \quad \sum_T Z_T(x)^3 = 0. \quad (2)$$

These are the Segre equations in Joubert coordinates. The equations and the outer S_6 -action are classical invariant theory [11]; the point used below is their realization as six amplitudes of one marked transfer.

The six coordinate lines obtained from either three-dimensional eigenspace form the classical real ETF(3, 6) associated with a symmetric conference matrix [12]. In measurement language these are the six icosahedral lines of the real three-level SIC [13]; they are not the nine-line informationally complete SIC for a complex qutrit. This distinction matters here because the transfer is real and its port orientation is part of the calibrated data.

3 Why the determinant retains orientation

The source construction identifies the determinant line and gives the basic plane-rotation counterexample for the permanent [2, Section 5]. We sharpen that observation by classifying the full and oriented double orbits in every dimension and proving that the determinant is the unique relative invariant of minimal degree.

Write $H = K^\dagger K$. For the intrinsic comparison, define

$$B_{\text{sym}}(K) = \text{tr}(\text{Sym}^3 H) = h_3(H), \quad F_{\text{ext}}(K) = \text{tr}\left(\bigwedge^3 H\right) = e_3(H) = \det H.$$

The dagger notation keeps the statement valid for the complex port-gauge extension; throughout the real Golden construction it is simply transpose. The first expression is the sum of all-transferred probabilities, averaged over the choice of normalized three-boson input occupation only after division by the ten-dimensional symmetric-cube input space. Thus the operational uniform-input average is $B_{\text{sym}}/10$. The second expression is the filled three-fermion probability.

For an orthogonal sign control, the singular values of K are the cosines of the principal angles between the output space and the controlled image of the input space. For a filtered control they remain transfer singular values, but need not be principal-angle cosines. The absolute determinant is their product. Its sign, after orientations are chosen, records whether the induced map between the two determinant lines preserves or reverses orientation.

Theorem 3.1 (Left–right orbit and orientation boundary). *Let V and W be real Euclidean spaces of the same dimension $n \geq 2$, and let $K : V \rightarrow W$.*

- (i) *A change of ordered orthonormal frames sends the matrix of K to*

$$K \mapsto R_-^\top K R_+, \quad R_\pm \in O(n).$$

The $O(W) \times O(V)$ orbit is determined by the singular values. If K is invertible, this orbit splits into two $SO(W) \times SO(V)$ orbits, distinguished by the sign of the oriented determinant. If K is singular, it does not split.

- (ii) *Intrinsically, $\bigwedge^n K$ is a map $\det V \rightarrow \det W$. Choosing orientations turns it into a scalar, and a frame change multiplies that scalar by $\det(R_-)\det(R_+)$. Thus its absolute value is an orthogonal invariant, while its sign is an invariant of oriented port spaces.*
- (iii) *The permanent of a matrix does not descend to either the orthogonal or the special-orthogonal double orbit. Hence it is a calibrated port-frame amplitude rather than a scalar attached to K as an unframed Euclidean map.*

Proof. The singular-value decomposition proves that two matrices lie in the same $O(W) \times O(V)$ orbit exactly when they have the same singular values. For an invertible map, special-orthogonal changes preserve the sign of the oriented determinant. Conversely, singular-value decompositions of two maps with the same singular values and determinant sign can be chosen with the same pair of orientation parities, so special-orthogonal left and right factors relate them. When a singular value is zero, reflecting its kernel direction changes one parity without changing the diagonal singular-value matrix. The two orientation classes therefore merge on the singular locus.

The functorial map on top exterior powers gives (ii), including its frame transformation law. For (iii), let R_θ be a plane rotation and use the special-orthogonal family $R_\theta \oplus I_{n-2}$. Along the right orbit of the identity matrix,

$$\text{per}(R_\theta \oplus I_{n-2}) = \cos(2\theta), \quad \det(R_\theta \oplus I_{n-2}) = 1.$$

The permanent therefore varies inside a single special-orthogonal double orbit. $\square$

Proposition 3.2 (The determinant is the minimal orientation carrier). *Up to scale, $\det K$ is the unique degree- n polynomial in the entries of K satisfying*

$$F(R_-^\top K R_+) = \det(R_-) \det(R_+) F(K) \quad (R_\pm \in O(n)).$$

In particular, an orientation-covariant amplitude of the lowest possible degree is forced to be the top exterior amplitude.

Proof. Apply diagonal sign changes on the left. Every monomial of a degree- n relative invariant must have odd degree in each of the n rows, hence degree one in every row. Right sign changes give degree one in every column. The polynomial is therefore a linear combination of the $n!$ permutation monomials. Invariance under orientation-preserving rotations in each pair of rows, followed by a reflection in one row, forces the coefficients of permutations differing by a transposition to have opposite signs. All coefficients are thus one common scalar times the permutation sign, which is the determinant expansion. $\square$

Every scalar invariant under the full orthogonal port action consequently factors through the singular values. This statement concerns the unframed map. A laboratory may freeze ordered phase-referenced port frames and then measure additional calibrated amplitudes.

Corollary 3.3 (Orientation chambers). *On the invertible real transfer space, the two oriented double orbits over a fixed singular-value point are exchanged by reversing one port orientation. They meet only on the determinantal hypersurface. Thus $\det K = 0$ is exactly the boundary at which the oriented top-exterior label disappears.*

Remark 3.4 (Complex port gauges). For complex port frames, $O(n)$ and $SO(n)$ are replaced by $U(n)$ and $SU(n)$. Singular values again classify the full double orbit. On the invertible locus, the residual label is the phase of the oriented determinant; on the singular locus it disappears. The real Golden transfer and a coherent phase reference reduce this determinant-line phase to the sign studied here.

The six-mode benchmark is the exceptional endpoint of a general conference classification, not the first member of a uniform family. Let $C = C^\top$ be a zero-diagonal sign matrix of order $2d$ with $C^2 = qI$, where $q = 2d - 1$. For a balanced half Y , write

$$C = \begin{pmatrix} A & R \\ R^\top & E \end{pmatrix}, \quad H_Y = Q_+^\top D_Y P_- D_Y Q_+,$$

where $P_\pm = (I \pm C/\sqrt{q})/2$, Q_+ is an isometry onto $\text{im } P_+$, and D_Y is the sign involution of the cut. Complementing the half gives $D_{Y^c} = -D_Y$ and hence $H_{Y^c} = H_Y$; all cut counts below are therefore projective.

Theorem 3.5 (Conference exchange rigidity). *For every symmetric conference matrix,*

$$\text{spec}(H_Y) = \text{spec}\left(\frac{1}{q} R R^\top\right) = \left\{1 - \frac{\alpha_i^2}{q} : 1 \leq i \leq d\right\},$$

where the α_i are the eigenvalues of A . This spectrum is independent of the balanced half exactly when $d \leq 3$. Hence order six is the unique nontrivial realized symmetric conference order with uniform balanced exchange spectrum. The order-two case is the trivial one-mode case, and no symmetric conference matrix of order four exists.

For a four-set K , let $w(K)$ be the sum of its three signed Hamilton-cycle products, call K aligned when $w(K) = 3$, and let c_Y count aligned four-sets contained in Y . Then

$$\mathrm{tr}(H_Y^2) = \frac{F_d + 32c_Y}{q^2}, \quad F_d = dq^2 - 2qd(d-1) + d(d-1) + 12\binom{d}{3} - 8\binom{d}{4}.$$

Hence the purity is cut-independent exactly in the same small orders.

Proof. Put $Q = C/\sqrt{q}$ and $L = [D_Y, Q]$. The commutator anticommutes with Q , and on the positive eigenspace $Lv = 2P_- D_Y v$. In cut coordinates,

$$L = \frac{2}{\sqrt{q}} \begin{pmatrix} 0 & R \\ -R^\top & 0 \end{pmatrix}.$$

Thus $4H_Y$ is L^*L on that eigenspace, and H_Y has the squared singular values of $R/\sqrt{q}$. Since the principal block of $C^2 = qI$ is $RR^\top = qI - A^2$, the spectral formula follows.

The identities $\mathrm{tr}(A^2) = d(d-1)$ and

$$\mathrm{tr}(A^4) = d(d-1) + 12\binom{d}{3} + 8 \sum_{\substack{K \subset Y \\ |K|=4}} w(K)$$

give the displayed purity formula, because $w(K)$ is 3 on aligned four-sets and -1 otherwise. Suppose $d \geq 4$ and this purity is independent of Y . The sums of the aligned-set indicator over all d -subsets are then constant. The characteristic-zero inclusion matrix from four-sets to d -sets has full column rank [10, Theorem 1], so the indicator is constant on four-sets.

Switch C so that every edge incident with one vertex is positive. For a four-set containing that vertex, if the other three edge signs are x, y, z , then $w = xy + xz + yz$. If the common value is 3, every triangle on the other $2d-1$ vertices is monochromatic; all its edges then have one sign, contradicting orthogonality of two conference rows. If the common value is -1 , the edge two-colouring has no monochromatic triangle, and $R(3, 3) = 6$ gives $2d-1 \leq 5$, again a contradiction.

For $d = 1$ the claim is immediate, and for $d = 2$ every two-by-two principal sign block squares to I . For $d = 3$, if τ is the product of the three edge signs, then $A^2 = 2I + \tau A$, so A^2 always has spectrum $\{4, 1, 1\}$. This proves both directions. $\square$

The higher-order residue is statistical rather than spectral. Greaves and Suda proved that the aligned sets form a $3-(2d, 4, (d-3)/2)$ design [8, Example 2.3]. Classical Johnson inclusion calculus [9] gives, for a uniformly chosen balanced half and $d \geq 4$,

$$\mathbb{E}c_Y = \rho \binom{d}{4}, \quad \mathrm{Var}(c_Y) = \frac{\binom{2d}{4} \binom{2d-8}{d-4}}{\binom{2d}{d}} \rho(1-\rho), \quad \rho = \frac{d-3}{2(2d-3)}.$$

Thus the purity has switching-class-independent mean and variance, obtained from these formulas by the affine change above; its possible values are quantized in steps of $32/q^2$. At order ten the aligned sets form a Steiner system and force the 126 projective cuts to split into 36 with $\mathrm{tr}(H_Y^2) = 145/81$ and 90 with $\mathrm{tr}(H_Y^2) = 59/27$. Paper III gives the same general theorem at the conference-operator stage [2, Theorem 5.1]; the local proof above keeps the present observable claim independently assessable. The aligned design and Johnson calculus are classical; the exchange interpretation and rigidity classification are the distinct layer used here.

Remark 3.6 (Collective inverse boundary). At order $2d \geq 10$, the complete labelled purity landscape recovers every aligned four-set by inclusion inversion. Paper III proves that this family determines the discrete signing up to switching and global negation [2, Theorem 5.2]; one calibrated triangle product selects the remaining bit relative to that calibration. This is a collective inverse corollary, not a local optical observable: the mean and variance alone are universal, and each individual H_Y remains orientation-blind.

Theorem 3.7 (Golden six-mode exchange benchmark). *Let $K_T(x)$ be a Golden transfer block. At every balanced control with three entries $+1$ and three entries -1 , for each of the six marked protocols,*

$$\text{spec } H = \left\{ \frac{1}{5}, \frac{4}{5}, \frac{4}{5} \right\}.$$

Consequently

$$B_{\text{sym}} = \frac{313}{125}, \quad F_{\text{ext}} = \frac{16}{125}, \quad B_{\text{sym}} - F_{\text{ext}} = \frac{297}{125}.$$

The corresponding uniform average over the ten normalized symmetric-cube input occupations is $B_{\text{sym}}/10 = 313/1250$. The mixed exchange sector has

$$\text{tr}(S_{(2,1)}(H)) = s_{(2,1)}(H) = \frac{8}{5},$$

and the Schur–Weyl checksum is

$$h_3(H) + 2s_{(2,1)}(H) + e_3(H) = \text{tr}(H)^3 = \frac{729}{125}.$$

In particular, every scalar invariant under the full orthogonal port action has the same value on all these balanced blocks and cannot distinguish their marked Segre nodes or orientations.

Proof. Order the coordinates by the negative support and its complement, and write

$$C_T = \begin{pmatrix} A & R \\ R^\top & E \end{pmatrix}.$$

If τ is the product of the three off-diagonal entries of A , then

$$A^2 = 2I + \tau A, \quad \det(tI - A) = t^3 - 3t - 2\tau.$$

Thus A^2 has spectrum $\{4, 1, 1\}$. The cross-Gram identity $RR^\top = 5I - A^2$ and Theorem 3.5 give the displayed spectrum of H for every balanced support, without a census. Equivalently,

$$\det(\lambda I - H) = \lambda^3 - \frac{9}{5}\lambda^2 + \frac{24}{25}\lambda - \frac{16}{125} = \left(\lambda - \frac{1}{5}\right) \left(\lambda - \frac{4}{5}\right)^2. \quad (3)$$

The complete and elementary symmetric functions then give

$$h_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{313}{125}, \quad e_3\left(\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right) = \frac{16}{125}.$$

Finally, the degree-three symmetric-function identity $h_3 - e_3 = p_1 p_2$ gives $297/125$. For the remaining partition of three,

$$s_{(2,1)} = \frac{p_1^3 - p_3}{3} = \frac{8}{5}.$$

Schur–Weyl decomposition

$$V^{\otimes 3} = \text{Sym}^3 V \oplus 2S_{(2,1)} V \oplus \bigwedge^3 V$$

then gives the checksum. The final assertion follows from Theorem 3.1. $\square$

The theorem does not say that calibrated bosonic measurements lose all control information. Once physical port axes are fixed, a permanent is an ordinary transition amplitude, accessible only through a phase-sensitive multiphoton protocol; photon counting measures its squared modulus. Such a value belongs to the calibrated apparatus rather than to the unmarked Golden operator. Likewise, the determinant sign belongs to the oriented amplitude: the probability F_{ext} does not retain it. The precursor below infers that sign from phase-referenced one-particle tomography; it does not measure a direct many-fermion phase.

4 Balanced controls and calibrated readout

The necessity of balance is independent of the Golden conference matrix.

Proposition 4.1 (Universal balance obstruction). *Let $V_+ \oplus V_-$ be an orthogonal splitting of $\mathbb{R}^{2d}$ into two d -planes. For a Boolean path control with negative support S , the cross block $K_S : V_+ \rightarrow V_-$ satisfies*

$$\text{rank } K_S \leq \min\{|S|, 2d - |S|\}.$$

Consequently a nonzero filled d -fermion amplitude is possible only for a balanced $d + d$ control.

Proof. Writing P_S for the coordinate projection onto the negative paths and using orthogonality of the two port spaces gives

$$K_S = -2Q_-^\top P_S Q_+,$$

so $\text{rank } K_S \leq |S|$. Complementing the control negates its cross block and gives the second bound. $\square$

There are 64 Boolean controls $x \in \{\pm 1\}^6$. The twenty balanced controls are the common nonzero determinant optimum. For each one,

$$|\det K_T| = \frac{4}{5\sqrt{5}}, \quad P_F = |\det K_T|^2 = \frac{16}{125}.$$

The other 44 controls have zero filled-fermion amplitude. This gives a sharp Boolean boundary rather than a numerical search criterion.

Proposition 4.1 makes every unbalanced mask rank at most two. There are $2\binom{6}{0} + 2\binom{6}{1} + 2\binom{6}{2} = 44$ such masks. The common balanced spectrum proves that all $\binom{6}{3} = 20$ remaining masks are invertible.

Freeze the port frames by a reproducible pivot convention. In that gauge the collision-free bosonic probabilities on balanced controls take the values

$$P_B(111 | 111) \in \left\{ \frac{16}{3125}, \frac{36}{3125}, \frac{64}{3125} \right\}.$$

These values are controls, not invariants. Coherent access to the calibrated amplitudes distinguishes all twenty oriented masks; squaring pairs antipodal controls and loses the overall orientation.

Across the ten projective balanced cuts, the six determinant sign words have pairwise Hamming distance six. Three selected signs identify a marked protocol once Golden membership is already known. Five signs and magnitudes jointly test membership and identify

the protocol. Measuring all ten signs supports soft matched-filter decoding; hard nearest-word decoding corrects two sign errors. No relative sign word fixes the global orientation without a coherent port reference.

For an explicit schedule, order the projective cuts lexicographically by the negative supports containing path zero:

$$012, 013, 014, 015, 023, 024, 025, 034, 035, 045.$$

The first three positions 012, 013, 014 classify the six protocols after membership is known. The five-cycle 012, 015, 023, 034, 045 jointly tests the common Golden magnitudes and classifies the protocol. These are fixed schedules for the transported marking above, not protocol-dependent choices.

Order the protocols by the one-line forms of their frozen path permutations. In the common calibrated orientation, the three-cut decoder is

protocol	T_0	T_1	T_2	T_3	T_4	T_5
P_T	012345	012354	012435	012453	012534	012543
signs	---	--+	-++	+--	++-	+++

The supplemental certificate records all ten signs for every marked protocol. Reversing the common orientation negates every signature; it does not change the relative classifier once that orientation is calibrated.

The universal complementarity identities for determinants and permanents of Jabbour and Cerf [6] provide a useful calibrated consistency check. They do not distinguish the Golden family by themselves.

5 Photonic design limit

Figure 1 separates three experiments that use the same one-particle transfer but answer different questions. Coherent light recovers a phase-referenced matrix and hence an inferred determinant sign. Ordinary three-photon input measures calibrated bosonic probabilities. Direct three-fermion emulation instead requires three identical copies of the transfer and an antisymmetric three-qutrit resource. The first two form a bounded precursor; the third is conditional on the additional source.

Circuit and port convention

Put $O_T = [Q_{T,+} \ Q_{T,-}]$, inject the first three eigenports numbered 0, 1, 2, and postselect the last three outputs numbered 3, 4, 5. Thus K_T is the row-3:5, column-0:2 block of U_T . For adjacent modes a, b , our compiled two-mode cell is

$$G_{ab}(\theta)|_{a,b} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

The $O(3) \times O(3)$ action of Theorem 3.1 is a gauge of the unframed Golden source object. Compiling O_T chooses physical, phase-referenced laboratory axes; a later port rotation changes that calibration or the device rather than acting as an unnoticed experimental gauge. In the frozen pivot gauge, input phases $S = \text{diag}(1, 1, 1, 1, 1, -1)$, followed in order by the cells below, reconstruct O_T . The table rounds angles to six decimal places and has maximum entry error below 7×10^{-7} . The machine-readable full-precision netlist in the supplemental certificate has error below 2×10^{-14} . Mode labels are zero-based and angles are in radians.

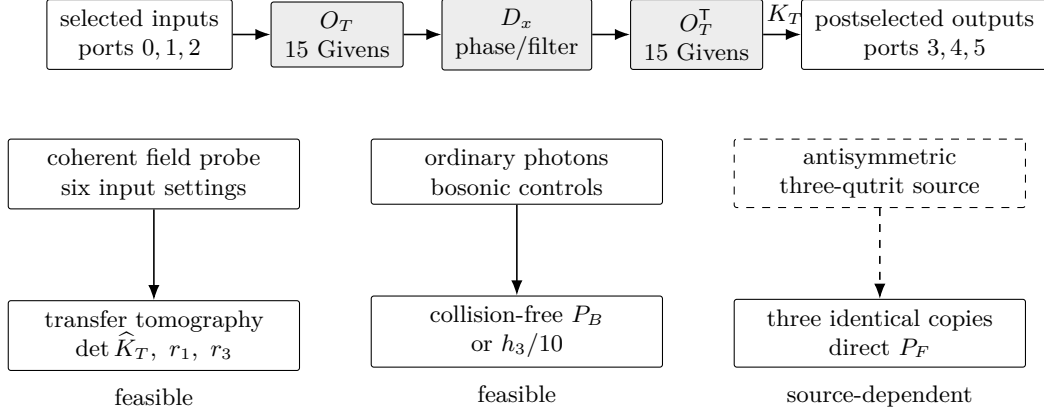


Figure 1: Circuit and readout boundary. The upper line implements $U_T(x) = O_T^\top D_x O_T$ in the filtered realization; a balanced mask may instead use the direct fifteen-cell compilation. The indicated subblock maps input ports 0, 1, 2 to output ports 3, 4, 5. Solid arrows are demonstrated measurement primitives. The dashed box and arrow mark the additional resource needed for direct three-fermion emulation. The structural nulls are $r_1 = \sum_T \hat{Z}_T$ and $r_3 = \sum_T \hat{Z}_T^3$; the drawing uses no color coding.

j	ab	θ_j	j	ab	θ_j	j	ab	θ_j
1	45	2.328295	6	45	-1.017222	11	01	-0.785398
2	34	-3.141593	7	12	-1.570796	12	12	-1.107149
3	45	2.328295	8	23	-2.356194	13	23	-1.047198
4	23	-1.570796	9	34	-0.955317	14	34	-0.955317
5	34	-1.570796	10	45	0	15	45	2.356194

A balanced $U_T(x)$, whose determinant is -1 , may be compiled directly into at most fifteen such cells plus one sign flip or π -phase element per copy. A general real filter uses the displayed sandwich with two fifteen-cell meshes and six variable attenuators; an amplitude transmission x_i means intensity transmission x_i^2 , with a 0 or π path phase supplying its sign. Three spatially parallel emulator copies therefore occupy eighteen modes and use at most 45 Givens cells plus three fixed π -phase elements in the balanced direct compilation. Encodings in time or frequency may reuse one physical mesh, but still require three matched logical copies. The other five marked protocols are obtained by the frozen path permutations; an odd representative exchanges the two port triples and transposes K_T .

Readout, nulls, and quantitative gates

Injecting each of the six inputs coherently and measuring the complex field at all outputs against one common local oscillator determines the transfer in six input settings, each with the phase sampling required by the detector. The common output reference is essential: intensity data and input-only interference leave independent output-row phases free and therefore cannot determine a block-determinant sign. The intensity/interference reconstruction of Rahimi-Keshari et al. [18] supplies the demonstrated transfer-characterization primitive; the determinant-sign claim here assumes the additional common input-output phase calibration. It is not a direct observation of a many-fermion phase. Programmable twelve-mode processors have separately operated with three photons [19].

The first acceptance test is structural, before comparison with any target probability.

For the six reconstructed amplitudes set

$$r_1 = \sum_T \hat{Z}_T, \quad r_3 = \sum_T \hat{Z}_T^3.$$

All six amplitudes must first be expressed in one calibrated gain and orientation convention. Independent normalization of each protocol would erase the content of these cross-protocol identities. For reporting across transmission levels, use the dimensionless scores

$$\rho_1 = \frac{|r_1|}{\sum_T |\hat{Z}_T|}, \quad \rho_3 = \frac{|r_3|}{\sum_T |\hat{Z}_T|^3},$$

whenever the respective denominator is nonzero. These are descriptive algebraic residuals, not goodness-of-fit statistics; experimental uncertainties must be propagated from the transfer reconstruction. Diagonal real loss changes x but leaves both nulls exact. Coherent diagonal phase error also leaves them exact over $\mathbb{C}$, although a literal real sign then needs the phase reference. Mesh error can move the amplitude word off the common Golden carrier: r_1 is the primary falsification test and r_3 the nonlinear confirmation. Passing both is necessary for carrier membership but does not certify the target transfer; the calibrated spectrum and probabilities supply the subsequent tests. Mismatch among the three emulator copies is a separate failure, because then no single block K_T describes the process. Each copy must be tomographed and matched before the determinant interpretation is applied. Uniform amplitude transmission λ scales amplitudes by λ^3 and probabilities by λ^6 , while ρ_1 and ρ_3 remain unchanged.

For two contraction blocks K and $K + E$,

$$|\det(K + E) - \det K| \leq 3\|E\|_2. \quad (4)$$

Indeed $A(t) = K + tE$ is a contraction for $0 \leq t \leq 1$, and

$$\left| \frac{d}{dt} \det A(t) \right| \leq \|\text{adj } A(t)\|_* \|E\|_2 \leq 3\|E\|_2,$$

because the three singular values of the adjugate are pairwise products of singular values of $A(t)$, hence at most one. Integration gives the bound. The balanced sign is certified by

$$\|E\|_2 < \frac{4}{15\sqrt{5}} = 0.119256\dots$$

This is an operator-norm requirement on transfer tomography, not a phase-shifter specification; experimental entrywise confidence regions must be propagated to this norm before the gate is applied.

The shot budget assumes independent Bernoulli accepted trials, ten-percent relative precision, a normal approximation, and simultaneous 95-percent coverage of all six probabilities. It excludes source preparation, coupling, propagation, and detector loss; these enter only when an accepted-trial rate is converted to laboratory time. With Bonferroni allocation over six two-sided intervals, put

$$z = \Phi^{-1} \left(1 - \frac{0.05}{12} \right), \quad n(p) = \left\lceil \frac{z^2(1-p)}{0.1^2 p} \right\rceil. \quad (5)$$

The table applies this formula to each ideal probability and rounds upward.

reading	probability	accepted trials	time at 100 s ⁻¹
balanced fermion, each protocol	16/125	4,742	0.79 min
weakest calibrated boson event	16/3125	135,250	22.5 min

The fermionic row is a conditional design budget; the feasible precursor cannot collect it without the antisymmetric source. At one accepted trial per second the corresponding times are 1.32 and 37.6 hours. The weakest calibrated bosonic control fixes the precursor shot budget. These are per-probability accepted-trial counts; a full run must multiply by the chosen protocols and cuts and separately budget phase samples, tomography repetitions, coupling loss, and detector acceptance.

External resource dependency

The emulator of Matthews et al. [14] sends an entangled resource through identical copies of a one-particle process. Here direct three-fermion emulation requires the totally antisymmetric state of three photonic qutrits. Goyal et al. [15] give a linear-optical preparation proposal, and Piccolini et al. [16] give a second interferometric proposal. To our knowledge, no experiment has prepared and characterized this three-photon qutrit singlet. The source is therefore an external dependency, not a component claimed here. A demonstrated three-qutrit GHZ state [17] has different exchange symmetry and neither supplies the resource nor predicts its engineering performance.

The value $P_F = \det H$ assumes normalized antisymmetric input and output states. In the emulator the six permutation amplitudes are combined coherently in one normalized antisymmetric output projection, so no factor of $3!$ multiplies the stated probability. Resource fidelity, photon indistinguishability, matching of the three transfer copies, and analyzer calibration are separate requirements.

Resource quality is more restrictive than shot noise under adversarial error models. If preparation noise is an arbitrary mixture with target-state weight η , keeping probability bias below ten percent for the balanced event requires $\eta \geq 0.9872$. With only a squared-Uhlmann-fidelity bound, denoted F_U , and the trace-distance worst case, the corresponding gate is $F_U \geq 0.999836$. A characterized noise model could relax these bounds, but the conclusion would then be model-dependent. For the mixture model, an event probability changes by at most $1 - \eta$, so ten-percent relative bias at ideal probability p requires $\eta \geq 1 - 0.1p$. Here $F_U = (\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}})^2$ is the squared Uhlmann fidelity. With fidelity alone, trace distance bounds the probability bias by $\sqrt{1 - F_U}$. Thus the two generic gates are

$$\eta \geq 1 - 0.1p \quad (\text{mixture model}), \quad F_U \geq 1 - 0.01p^2 \quad (\text{fidelity only}). \quad (6)$$

Substitution of $p = 16/125$ gives the two displayed balanced-event gates.

6 Discussion

The Golden transfer separates three levels of information. Its singular values determine every intrinsic bosonic scalar. A calibrated port frame adds permanent amplitudes and oriented control information. Antisymmetric exchange supplies the determinant as a one-dimensional, orientation-covariant amplitude of the port-frame action; its event probability discards the sign. The proposed precursor recovers the signed matrix element by coherent one-particle tomography after the laboratory frames have been compiled and oriented. The Segre coordinate is therefore not an arbitrary cubic observable, but neither is its sign claimed here as a direct many-fermion measurement.

The six-mode benchmark is also the only nontrivial symmetric conference case in which every balanced control has the same singular spectrum. Higher orders retain a universal statistical envelope—the exact purity mean and variance—but not a universal spectrum.

This distinction is operational: singular statistics remain orientation-blind, while the determinant signs used by the calibrated decoder carry the missing relative orientation.

The example also separates a feasible optical certification from a direct many-particle realization. Current mesh and tomography technology can test the one-particle carrier and its bosonic shadows. Direct photonic emulation of the three-fermion amplitude depends on an additional antisymmetric resource. This boundary is part of the result rather than an experimental claim.

Data availability

No empirical data were generated or analyzed. The exact mathematical artifacts supporting the finite claims—the standard-library checker, compact certificate, SHA-256 manifest, locked symbolic environment, and independent replay entry points—are included with the article’s supplemental package. The submitted artifact version is identified in that package; live hosting and its public locator are supplied with the publication record.

Verification

Human proofs cover the Boolean rank boundary, general rigidity, local balanced spectrum, symmetric functions, orbit theorem, and decoder radius. Classical citations supply the design statistics; Paper III supplies labelled reconstruction. The local certificate does not reprove either result. It independently checks all twenty balanced masks, the three calibrated permanent values, six sign words, full-precision netlist, and fifteen-cell compilation. Their three atomic source bundles have exact generators and independent replays; one paper-local manifest pins every load-bearing file.

From the supplemental-package directory, **make check** verifies source hashes and independently recomputes every imported rational identity, census total, Hamming distance, schedule, and source replay used above. This evidence does not certify experimental feasibility, source availability, or literature priority.

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